MARKS ARE AWARDED STRICTLY AS PER THE ANSWER KEY ONLY

MARKS MAY INCREASE OR DECREASE AFTER REVALUATION. THIS WILL BE FINAL

DO NOT WRITE PERSONAL REQUESTS!

Exam Date & Time: 24-Nov-2025 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
SUBJECT: BIO 1071/BIO_1071 BIOLOGY FOR ENGINEERS

Marks: 50 Duration: 180 mins.

1A

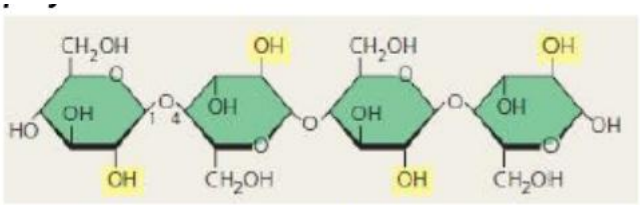
Sl no	Criterion	Explanation	Marks
1	Bonding ability	Silicon can from bonds similar to carbon because of valency 4	1
2	Catenation,smaller size, low weight	Makes a perfect building block	1
3	Abundancy	Silicon is abundant in space dust	1

1B

Protein Structure Level	Stabilizing Forces	Susceptibility to Denaturation	Explanation
Secondary	Hydrogen bonds between backbone atoms (e.g., α-helices, β-sheets)	High	Disrupted by heat or pH, but relatively stable due to repetitive hydrogen bonding. 1 mark
Tertiary	Hydrophobic interactions, ionic bonds, hydrogen bonds, disulfide bridges (3D folding)	High	Complex folding relies on weak interactions that are easily disrupted by denaturing agents. 1 mark

Less affected structure: Primary structure because it involves strong covalent peptide bonds 1 mark

1C

(a) 

2 marks

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b) Strutural feature: **Cellulose: 1–4 linkage of β glucose monomers.** In cellulose, every β glucose monomer is upside down with respect to its neighbors. **(1 mark)**

Functional feature: Form the mechanical support to the cell, forms energy source to microbes (1 mark)

2A

Punnette square 2 marks

	PT	Pt	pT	pt
pt	PpTt	Pptt	ppTt	pptt
pt	PpTt	Pptt	ppTt	pptt
pt	PpTt	Pptt	ppTt	pptt
pt	PpTt	Pptt	ppTt	pptt

Genotype	Phenotype
PpTt	Purple, Tall
Pptt	Purple, Short
ppTt	White, Tall
pptt	White, Short

1 mark

2B

Total offspring = 965 + 944 + 206 + 185 = 2300

Number of recombinants = 206 + 185 = 391

Recombination frequency:

$$RF = \frac{391}{2300} \times 100\% \approx 17\%$$

(a) **2 marks**

(b) A recombination frequency of 17% indicates the two factors are linked on the same chromosome and are approximately 17 map units (cM) apart 1 mark

2C

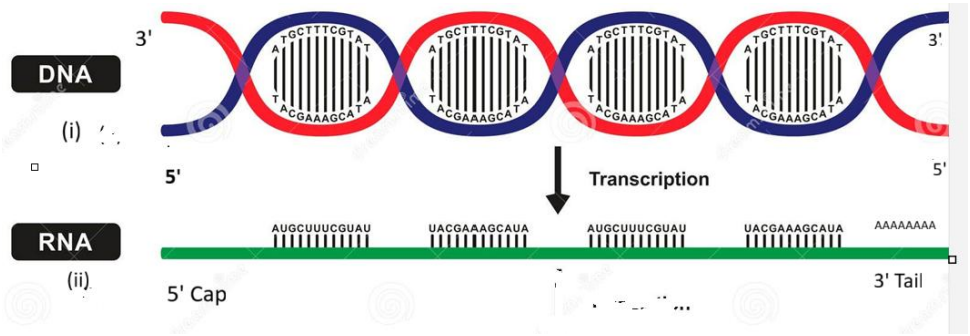
A. **Autosomal dominant**
B. **The trait appears in every generation, which is typical of dominant inheritance. Both males and females are affected → the trait is autosomal, not sex-linked, affected father transmits the trait to both son and daughter hence it cant be x linked dominant.**
C. **Both Aa type**

3A

Final Proportions After 3 Generations

- Heavy (HH): 0%
- Hybrid (HL): 25%
- Light (LL): 75%

3B



(iii) Corresponding stop codons in DNA:

5'TAA3' or 5'TAG3' or 5'TGA3'

3'ATT5' or 3'ATC5' or 3'ACT5'

Any one 1 mark

(iv) Second amino acid is Leucine, hence tRNA code for 5'CUU3' is 3'GAA5' (or written 5'AAG3') 1 mark

3C

Answer (3 Marks – 3 Points):

1. **Role of ^{35}S:** Labels **protein** coat, since sulfur is found in amino acids (methionine, cysteine) but not in DNA.
2. **Role of ^{32}P:** Labels **DNA**, since phosphorus is present in DNA (phosphate groups) but not in proteins.
3. **Conclusion (Cake Analogy):** If Hershey and Chase were baking a "DNA cake," they'd label the **phosphorus (^{32}P)** — the "cake" itself — to prove the genetic material is DNA, not just the "frosting" (protein).

4A

(i) predicted long-tongued moth → later confirmed → evidence of coevolution

(ii)

- +/+ mutualism (both benefit)
- +/- predation/parasitism (one benefits, one harmed)
- -/- competition (both harmed)

4B

Respiratory efficiency – Nostrils shifting position improves breathing in new environments.
Habitat specialization – Drift reflects adaptation to aquatic vs. terrestrial life, aligning nostril placement with lifestyle needs.
Evolutionary transition – Demonstrates gradual structural change over time, supporting descent with modification and adaptive evolution.

4C

Create new traits (Eg. Antibiotic resistance)
Adaptive advantage (Eg Colour of insects)
Structural (Eg Body design of neck of giraffs)
Functional improvement (Eg Sickle trait gives malarial resistance)
Consider any other relevant examples

5A

Consider any three

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#	Natural Model	Human Application	Key Concept	Benefit
1	**Lotus leaf**	Self-cleaning paints & surfaces	**Superhydrophobicity**	Dirt/water repelled, low maintenance
2	**Kingfisher beak**	Bullet train nose design	**Streamlined aerodynamics**	Reduced noise, improved speed
3	**Shark skin**	Anti-bacterial hospital surfaces, swimsuits	**Micro-ribbed texture**	Reduced drag, prevents microbial growth
4	**Spider silk**	Strong lightweight fibers	**High tensile strength**	Durable, flexible materials
5	**Termite mounds**	Passive cooling in buildings	**Natural ventilation**	Energy-efficient temperature regulation

5B

#	Adaptive Feature	Biomimicry Concept
1	**Water-channeling skin grooves**	**Passive water collection**
2	**Camouflage coloration**	**Adaptive concealment**
3	**Spiny body armor**	**Protective design**
4	**Fat storage in tail**	**Energy reserve system**
5	**Low metabolic rate**	**Energy efficiency**

Consider any three

5C

Consider the following concepts any four

#	Living Organism	Technology Inspired	Key Concept	Benefit
1	**Kingfisher beak**	Bullet train nose design	**Streamlined shape**	Reduced air resistance
2	**Kingfisher diving ability**	Train aerodynamics	**Silent entry into water**	Lower tunnel boom noise
3	**Beak tapering**	Nose cone structure	**Gradual pressure change**	Smooth airflow transition
4	**Natural efficiency**	Energy use in trains	**Minimal splash energy loss**	Improved fuel efficiency
5	**Precision hunting**	Engineering design	**Accurate targeting of prey**	Enhanced speed and stability